TEXORA · LECTURE SHEET

Machine Tools: Lathe, Drill, Grinder, Shaper, Saw and Milling

The six machines of the workshop and their governing arithmetic: cutting speed V = πDN/1000 everywhere; lathe operations, drilling family, grinding wheels, the shaper's quick return, power sawing, and milling with its feed calculations and up/down choice.

ME-1208 · ENGINEERING WORKSHOP · TOPIC 2 · LEVEL 1 TERM 2

1. One formula rules them all

Every rotating tool obeys the same cutting-speed law — surface speed of the work (lathe) or tool (drill, mill, wheel):

$$V = \pi D N/1000 \quad \text{m/min} \quad (D \text{ in mm, } N \text{ in rev/min}) \text{ — inverted: } N = 1000\,V/(\pi D)$$

WORKED EXAMPLES

Turning a Ø50 mm bar at 400 rev/min: V = π × 50 × 400/1,000 = **62.8 m/min**.
Recommended V for mild steel with HSS ≈ 30 m/min on a Ø40 mm job: N = 30,000/(π × 40) = **239 rev/min** — choose the nearest lower machine speed.
Machining time for a 150 mm turning pass at feed 0.2 mm/rev, 300 rev/min: T = L/(f N) = 150/(0.2 × 300) = **2.5 min**.

2. The lathe — the mother machine

The work rotates between **centres** or in a **chuck**; the single-point tool, carried by the carriage (saddle, cross-slide, compound rest), feeds along or across. Main parts: bed, headstock (spindle + speed gearbox), tailstock, lead screw (threading) and feed rod. Operations: **facing** (flat end), **plain/parallel turning**, **step turning**, **taper turning** (compound-rest swivel for short tapers, tailstock offset, taper attachment), **boring** (enlarging a hole), **drilling/reaming** from the tailstock, **knurling** (diamond pattern grip), **parting off**, and **thread cutting** — the lead screw geared so the tool advances exactly one pitch per spindle revolution. Work-holding: 3-jaw self-centring chuck (fast, round stock), 4-jaw independent chuck (exact centring, odd shapes), faceplate, collets, between centres with a carrier for long accurate shafts.

3. Drilling machines

The drill rotates and feeds into still work (bench, pillar and radial-arm sizes). The operation family: **drilling** (new hole, twist drill), **reaming** (size and finish an existing hole), **boring**, **counterboring** (flat-bottomed enlargement for bolt heads), **countersinking** (conical seat for flat-head screws), **spot facing**, and **tapping** (internal threads with a tap). Speed comes from the same formula — a Ø10 mm drill for 25 m/min wants N = 25,000/(π × 10) ≈ 800 rev/min — and small drills need the highest speeds, which is why drilling spindles are geared fast. The drill's two flutes clear chips and admit coolant; a wandering start is prevented by the centre punch of Topic 1.

4. Grinding

Grinding finishes with thousands of tiny cutting edges — abrasive grains (aluminium oxide for steels, silicon carbide for cast iron and carbides) bonded into a wheel. **Surface grinders** flatten; **cylindrical grinders** finish shafts between centres; tool-and-cutter grinders sharpen. Wheel specification is stamped on the blotter (abrasive, grit size, grade, structure, bond) and its **maximum safe speed must exceed the spindle speed** — a Ø200 mm wheel at 3,000 rev/min runs V = π × 0.2 × 3,000/60 = 31.4 m/s, near the usual 33 m/s class limit. Safety liturgy: ring-test the wheel (a clear ring when tapped = sound; a dull thud = cracked — destroy it), use the guard, stand aside at start-up, dress the wheel with a diamond dresser when it loads or glazes.

5. Shaper and power saws

The **shaper** reciprocates a single-point tool over still work, cutting only on the forward stroke; its crank drive gives a **quick-return** (the return stroke is about twice as fast — typically a 2:1 ratio), because idle time is wasted time. It makes flat surfaces, keyways and slots on modest work cheaply, and the planer is its big brother (tool still, table moves). **Power saws** mechanise Topic 1's hand sawing: the **power hacksaw** (reciprocating blade, cuts on one stroke), the continuous **band saw** (fast, versatile, also contours) and the **circular saw**; blade pitch follows the same three-teeth-in-contact rule.

6. Milling

The milling cutter rotates with many teeth; the table feeds the work past it in three axes. Families: **slab/peripheral milling** (axis parallel to the surface), **face milling** (axis perpendicular — the modern production default), **end milling** (slots, profiles, pockets with an end mill) on vertical or horizontal machines, plus gear cutting with form cutters on a dividing head. Feed is set per tooth:

$$\text{Table feed} = f_z \times z \times N \quad \text{(mm/min)}$$

Example: f_z = 0.1 mm/tooth, 6 teeth, 500 rev/min → 0.1 × 6 × 500 = **300 mm/min**. **Up (conventional) milling**: cutter teeth bite against the feed — chip grows from zero, tool rubs then bites; safe for old machines with backlash and scaly surfaces. **Down (climb) milling**: teeth bite with the feed, starting at full chip — better finish and tool life, but it pulls the work, so it demands a backlash-free (CNC) machine.

7. Key points

- V = πDN/1000 and N = 1000V/(πD) run every rotating machine; time = length/(feed per rev × revs).
- Lathe: work rotates; chuck vs centres; threading is lead-screw synchronisation, one pitch per revolution.
- Drilling family: drill → ream (finish) → counterbore/countersink (seats) → tap (threads).
- Grinding: never exceed the wheel's marked speed; ring-test before mounting; Al_2O_3 for steel, SiC for cast iron/carbide.
- Shaper cuts one way with ≈ 2:1 quick return; milling feed = f_z × z × N; climb milling needs a backlash-free machine.

8. Practice

Answer in your own words; brief model answers follow.

1. A Ø60 mm shaft is to be turned with HSS at a recommended cutting speed of 25 m/min. Find the spindle speed to set, and the time for one 200 mm pass at 0.25 mm/rev feed.
2. A face mill has 8 teeth, runs at 400 rev/min, and is fed at 0.08 mm/tooth. Find the table feed in mm/min.
3. State two differences between up milling and down milling, and why down milling is barred on an old machine with table backlash.
4. A grinding wheel marked "max 33 m/s" is Ø250 mm. Find its maximum permitted spindle speed and check whether 2,500 rev/min is legal.
5. Which lathe work-holding device for each: (a) a batch of round bar blanks, speed matters; (b) an eccentric square casting that must be centred to 0.02 mm; (c) a long shaft to be turned along its whole length?
6. Why does the shaper waste less time than a simple reciprocating saw of the same stroke, although it also cuts on only one stroke?

Answer key

1. N = 1,000 × 25/(π × 60) = 25,000/188.5 = 133 rev/min (set 130-ish, the nearest available step not exceeding it). Feed rate = 0.25 × 133 = 33.3 mm/min; time = 200/33.3 ≈ 6.0 min.
2. Feed = 0.08 × 8 × 400 = 256 mm/min.
3. Chip thickness: up milling starts at zero and grows (initial rubbing, work-hardening of the surface); down milling starts at maximum and ends at zero (clean shearing, better finish). Force direction: up milling lifts the work and pushes against the feed screw steadily; down milling pulls the work into the cutter. On a machine with backlash the pull snatches the table forward by the slack amount each tooth-bite — chatter, oversize cuts, broken teeth. Backlash-free ball-screw CNC machines remove the hazard.
4. Wheel surface speed V = πDN/60 (m/s with D in m): N_max = 33 × 60/(π × 0.25) = 1,980/0.785 = 2,522 rev/min. At 2,500 rev/min the actual speed is π × 0.25 × 2,500/60 = 32.7 m/s — inside the limit, legal but with slim margin; any faster spindle setting would not be.
5. (a) 3-jaw self-centring chuck — grips and centres round stock in seconds. (b) 4-jaw independent chuck — each jaw adjusted separately with a dial indicator to centre an irregular part exactly. (c) Between centres with a lathe carrier (dog) — the whole length stays accessible, and the shaft can be reversed end-for-end while keeping the same axis.
6. Because of its quick-return mechanism: the crank geometry makes the return stroke about twice as fast as the cutting stroke, so the idle half of the cycle consumes a third of the time instead of half — the same one-way cutting, a shorter penalty. (The power hacksaw uses the same idea.)

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